Algebra 1
Unit 4
Lesson 10 Practice

Name Answer Key___________
Date____________________
Period___________________

1. Write the equation of the line that passes through $\overset{x}{(-2},\overset{y}{1})$ and is parallel to $y = 2x - 3$ in point-slope form.

$m = 2$

$y - y_1 = m(x - x_1)$

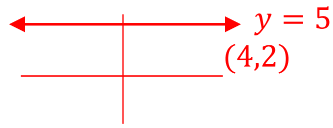
$\boxed{y - 1 = 2(x + 2)}$

2. Write the equation of the line that is parallel to $y = 5$ and passes through point $(4, 2)$.

horizontal line

$m = 0$

$\boxed{y = 2}$



3. Write the equation of the line that contains point $(\overset{}{0}, -3)$ and is parallel to $4x + 3y = 6$ in slope-intercept form.

$x \quad y$

$-4x \qquad -4x$

$\frac{3y}{3} = \frac{6}{3} - \frac{4x}{3}$

$y = 2 - \frac{4}{3x}$

$m = -\frac{4}{3}$

$(0, -3)$

$\downarrow$

y-intercept

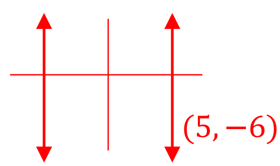
$\boxed{y = -\frac{4}{3}x - 3}$

4. Write the equation of the line that is parallel to $x + 6 = 0$ and passes through point $(5, -6)$.

$\underline{\quad -6 \quad -6}$

$x = -6$ (vertical line)

$\boxed{x = 5}$



5. Write the equation of the line that passes through point $(3, -4)$ and is parallel to $y + \frac{1}{7} = -\frac{8}{7}(x - 1)$ in standard form.

$-\frac{8}{7}x$

Slope= m

$-\frac{8}{7}$

$y + 4 = -\frac{8}{7}(x - 3)$

$y(7) + 4(7) = \frac{-8(7)}{7}x + \frac{24(7)}{7}$

$7y + 28 = -8x + 24$

$+8x \qquad\qquad + 8x$

$7y + 8x + 28 = 24$

$\qquad -28 \quad -28$

$\boxed{8x + 7y = -4}$

6. Write the equation of the line, in all three forms, that passes through point $(-4, 11)$ and is parallel to $y + 8 = -\frac{5}{3}(x - 3)$. $-\frac{5}{3} =$ slope

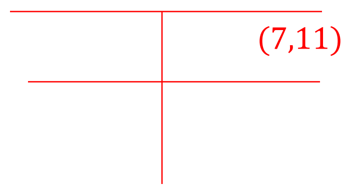
$x \quad y$

Point-Slope Form	Slope-Intercept Form	Standard Form
$y - 11 = -\frac{5}{3}(x + 4)$	$y - 11 = -\frac{5}{3}x - \frac{20}{3}$ $+11 \qquad + 11$ $y = -\frac{5}{3}x + \frac{13}{3}$	$(3)y = -\frac{5(3)}{3}x + \frac{13}{3}(3)$ $3y = -5x + 13$ $\underline{+5x \quad +5x}$ $5x + 3y = 13$

horizontal

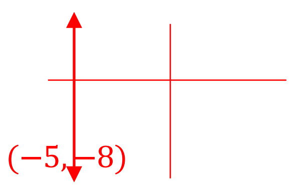
7. Write the equation of a line that is parallel to $y = 0$ and contains point $(7, 11)$.

$\boxed{y = 11}$



$x =$ vertical

8. Line K passes through $(-5, -8)$ and is parallel Line R which has an undefined slope. Write the equation of Line K.



$\boxed{x = -5}$

9. Write the equation of the line, in all three forms, that is parallel to the line that passes through the points $(10, 1)$ and $(6, -1)$ and passes through point $(8, 3)$.

x	y
10	1
6	-1

$-4 \qquad -2$

$\frac{\Delta y}{\Delta x} = \frac{-2}{-4} = \frac{1}{2}$

Point-Slope Form	Slope-Intercept Form	Standard Form
$\boxed{y - 3 = \frac{1}{2}(x - 8)}$	$y - 3 = \frac{1}{2}(x - 8)$ $y - 3 = \frac{1}{2}x - 4$ $\underline{+3 \qquad + 3}$ $\boxed{y = \frac{1}{2}x - 1}$	$(2)y = \frac{1(2)}{2}x - 1(2)$ $2y = x - 2$ $\underline{-x \quad -x}$ $\boxed{-x + 2y = -2}$

10. Write the equation of the line, in all three forms, that is parallel to the line that passes through the points $(2, 7)$ and $(-1, 17)$ and passes through point $(2, 20)$.

	x	y	
-3	2	7	$+10$
	-1	17	

$m = \frac{\Delta y}{\Delta x} = \frac{10}{-3}$

Point-Slope Form	Slope-Intercept Form	Standard Form
$\boxed{y - 20 = -\frac{10}{3}(x - 2)}$	$y - 20 = \frac{-10}{3}(x - 2)$ $y - 20 = \frac{-10}{3}x + \frac{20}{3}$ $\boxed{y = -\frac{10}{3}x + \frac{80}{3}}$	$(3)y = \frac{-10(3)}{3}x + \frac{80(3)}{3}$ $3y = -10x + 80$ $\underline{+10x \quad +10x}$ $\boxed{10x + 3y = 80}$